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## Lab5: “MCCNT underflow Interrupt”

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### Introduction and objectives

The HCS12 microcontrollers feature a Timer Module that can be configured to generate interrupts upon specific timing events, including overflow, compare, and underflow conditions.

In this lab session, you will explore the use of timer-based interrupt through the modulus down-counter mechanism, where the MCCNT timer counts down from a preset value to zero, then automatically reloads and continues counting. An **underflow interrupt** is triggered each time the counter reaches 0 , allowing precise control over timed events without constant CPU supervision.

Through this hands-on session, you will be able to configure, enable and monitor an internal maskable interrupt and you will understand how to create periodic delays and their use in real-time embedded applications.

### Pre Lab5 :

In this preliminary task, you are required to answer the following questions :

#### ExI:

- Define the internal and external interrupts and give an example of each type.
- Describe in details how do these processors handle interrupts. (Program Counter, Stack, Vector Address, ISR ..).

- c) What is the Modulus Count Down Counter MCCNT?  
Is it 16 or 8 bits register? Deduce the maximum value it will start counting down with .
- d) When does it trigger an interrupt?
- e) By referring to the “HCS12” datasheet (located in folder Microcontroller), explain the function of the following registers and the role of each underlined bit:
- MCCTL register (MCZI, MODMC, MCEN, MCPRI, MCPRI bits)
  - MCFLG register (MCZF)
  - MCCNT
- f) What is the value of the MCCTL register?
- g) what is the value of the MCFLG register?
- h) Get the Vector Address of the Modulus down counter underflow interrupt from the interrupt vector locations table(refer to “HCS12” datasheet). (*This address will be placed at the end of your code*)

### Lab5: “Interrupt”

Write fully commented programs for the PBMUSLK board, including appropriate labels, memory operands and constants, to solve the 3 tasks below:

#### Task 1:

Using MCCNT underflow interrupt, the first program should do the following:

- a) Toggle ON/OFF the 4 LEDs periodically every 200ms.
- b) Toggle the LEDs every 2s.
- c) Toggle LED1 every 2s and LED2 every 4s.

Hint: - Use same ISR (load the same value in MCCNT) - Use variables to help you modify the delay time needed to turn ON/OFF a LED.

Task 2:

The second program should do the following:

Blink the LEDs every 500ms using the MCCNT underflow interrupt. Using IRQ interrupt, the LEDs will stop blinking whenever PB1 is pressed and stay OFF till the next press, the LEDs will restart blinking and the cycle will be repeated continuously.

Task 3:

The third program should do the following:

Shift the LEDs to the left every 500ms then shift to the right with the same time delay using MCCNT underflow interrupt.

Using IRQ interrupt:

- a) the LEDs are turned OFF when PB1 is pressed the first time.
- b) the LEDs are turned ON when PB1 is pressed the second time.
- c) the shifting cycle will restart from the beginning when PB1 is pressed the third time.

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